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REVIEW ARTICLE

Facile fractionation of lignocellulosic biomass: a review on promising green technology of deep eutectic solvent for bioethanol production

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Abstract

Sustainable bioethanol production from lignocellulosic biomass to meet the energy demand and generate wealth from waste is the need of the hour. A surplus amount of lignocellulosic biomass is produced worldwide as organic waste which can be utilized for bioethanol production. Pretreatment is a crucial step in biorefinery application, the selection of the best pretreatment technology will reduce the intensive energy requirements, and the operational costs make biofuel production economically feasible and sustainable. Every pretreatment technology has its own drawbacks which needs to be addressed. Among these technologies, deep eutectic solvent (DES) is being investigated as an ideal solvent for the pretreatment of lignocellulosic biomass to improve the enzymatic digestibility, surface area solubility, and lowering lignin content and cellulose crystallinity. DES offers a green way of deconstructing lignocellulosic biomass with minimum environmental hazard. DES is an inexpensive and environment-friendly solvent effective in biomass valorization. This review focuses on the current status and advancements in pretreatment technologies using DES and its future prospects.

Keywords Lignocellulosic biomass · Pretreatment · Green solvent · Delignification · Deep eutectic solvent (DES) · Bioethanol

1 Introduction

The constant consumption of fossil fuels has various socio-economic impact including climate concerns such as acid rain, melting glaciers, and global warming. It warrants immediate attention towards the carbon–neutral and low-emission forms of renewable biofuel to reduce the dependency on fossil fuel [1–3]. The lignocellulosic biomass, an alternate of finite energy source, is abundantly available in nature. It has been assessed that approximately 100 billion tons of biomass per year are produced globally. According to the International Energy Agency, bioenergy can provide approximately 10% of the primary energy supply worldwide by 2035, and biofuel can replace up to 27% of transportation fuel by 2050 [4]. Due to the excessive energy density of bioethanol and greater heat vaporization, it is considered the best potential biofuel to replace fossil fuel [5, 6]. The cost of raw material used for ethanol production is a major bottleneck in the process of biofuel production which can be alternatively reduced by using a cheap source of feedstock such as lignocellulosic biomass [7]. Lignocellulosic feedstocks can be categorized as crop residue such as sugarcane bagasse, rice husk, rice straw, and wheat straw, wooden residue such as wood dust and wood chips, herbaceous biomass such as pomace and door grass, and forest residue. Globally, the crops like rice, maize, and wheat are major crops that are considered as a source of lignocellulosic biomass (LCB) and used to produce dry biomass of about 5300 million tons per year [8–10]. Lignocellulose is

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rich carbohydrate source, and it consists of lignin, cellulose, and hemicellulose that are chemically linked through covalent and noncovalent forces. Last few decades have witnessed that these monomers can be converted in bioethanol through microbial fermentation process [11]. Biofuels can be categorized into first, second, third, and fourth generation of biofuel based on the biomass utilized in the production, First-generation biofuel is mainly produced from edible food such as sugarcane (as in Brazil) and corn (as in the USA). Second-generation bioethanol is produced from renewable lignocellulosic biomass such as crop residues and agriculture and allied industrial byproducts [12, 13]. The third generation of biofuel is mainly produced from aquatic biomass such as microalgae and cyanobacteria that can be used to biodiesel or any other high-energy biofuel. However, the fourth generation of biofuels is derived from genetically engineered microorganisms including microalgae, yeast fungus, and cyanobacteria [14].

Over the years, lignocellulosic biomass has gained significant attention as a wide range of feedstock that can be utilized as the substrate for biofuel production without any competition with food [12–15]. Lignin is covalently linked to carbohydrates, viz., cellulose and hemicellulose, which make it difficult to access by the enzyme. In order to fully utilize the carbohydrate, lignin is a physical barrier that needs to be removed [16]. Therefore, a pretreatment process is needed to make the structural changes in the biomass to make it more accessible to enzymes that convert the polymer of carbohydrates into fermentable sugars [17]. There are a variety of pretreatment methods that are used to increase the digestibility of lignocellulosic biomass; these include mechanical pretreatment, chemical pretreatment, biological pretreatment, and many more [18]. All pretreatment processes have many advantages and disadvantages. In general, the disadvantages are related to high energy consumption, high cost of consumables, negative environmental impact, toxicity, and slow rate of conversion. Physical pretreatment process such as microwave or ultrasonic pretreatment process are energy-consuming [19]. Although acidic and alkali pretreatment processes efficiently hydrolyze cellulose and hemicellulose, low sugar yield and high corrosive nature of chemicals remain a bottleneck in adopting this technology at industrial scale. On the other hand, some pretreatments like organosolv, ozonolysis, steam explosion, ammonia fiber explosion (AFEX), and liquid hot water are harnessing the physiochemical nature of their method. These physiochemical pretreatment processes utilize power, directly or indirectly, which makes it high energy-consuming. The energy lost in such pretreatment processes cannot be recycled and hence are considered as energy utilizing instead of energy-producing technology[20].

Ionic liquid solvent (ILs) is a mixture of two salts organic cation and inorganic anion. The pretreatment of biomass with ILs forms a complex of electron donor and acceptor ions with the cellulose and helps in efficient solubilization of the cellulose [21]. Deep eutectic solvent (DES) is a solvent with special properties, formed by mixture of various ionic molecules either anion or cation of Lewis or Brønsted acid/base. The mixture of hydrogen bond acceptor and hydrogen bond donor leads to unusual physical properties including high depression melting point [22]. It is noteworthy that the physical properties of deep eutectic solvents are similar to ILs, but their chemical properties are significantly different. DES comprises mixture of large, nonsymmetrical ions having low lattice energy and lower melting point; however, ILs are pure compounds. Therefore, DES is considered as the new class of ILs [21]. The general formula for DES could be described as ($Cat^+ X \rightarrow zY$), where Cat^+ is cation, *X* is Lewis base, *Y* is either Lewis or Brønsted acid, while *z* refers to the number of interacting *Y* molecules [23]. By mixing the hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD), the melting point of the mixture decreases due to the charge delocalization [24]. Choline chloride (ChCl) or choline is the most studied anionic part for the pretreatment of various types of biomasses. The low melting point of DES is because of strong hydrogen bond interactions between HBD and HBA and due to the presence of an anionic part of ChCl [25]. Several chemicals are used as hydrogen bond donors to create a ChCl-based DES system, viz., amines, amides, polyalcohols, polysaccharides, monocarboxylic acids, dicarboxylic acids, and other acids [25]. Due to their unique physical properties, an escalation in the research involving DES as solvent has been witnessed during last few years. Researchers found that a large amount of lignin is solubilized by the DES with the minimum effect on cellulose. This property of DES makes it a suitable candidate for lignin extraction and removal from lignocellulosic biomass [26]. Lately, DES has been acknowledged as an alternative to ILs because of its biocompatibility with biomolecules, i.e., enzymes, proteins, nucleic acid, and microbes. DES has attracted interest in various applications such as in bio-pharma industries as bio-organic catalysts, molecular extractions, and biotransformation [27]. The unique ability of DES as a solvent for biomolecules including carbohydrates and its polymer (such as xylose, arabinose, starch, cellulose, and chitin) and lignin makes it an interesting field of research for the global scientific community to value-added products such as biofuel [7].

2 Basic composition of lignocellulosic feedstock

The main constituents of lignocellulose biomass are cellulose (9–80%), hemicellulose (10–50%), and lignin (5–35%) which forms a network of polymers**.** According to an estimate, 70% of total biomass is composed of carbohydrate

and lignin polymer [28], while the residual fraction (30%) includes resins, terpenoids, proteins, and ash. However, the selection of the best pretreatment method is largely dependent on the physico-chemical property of biomass. Table 1 shows different lignocellulosic feedstock and their composition of the cellulose, hemicellulose, and lignin content. Cellulose, the most abundant biopolymer, constituted into different amorphous, and crystalline, regions, represents approximately 40–60% of the weight. It consists of a repetitive unit of ß-D-glucopyranose held together via β-(1,4) glycosidic linkage. Cellulose microfibrils are formed by the tight arrangement of 500–1400 D-glucose units [29]. By cellulose modification, reactive -OH groups form a number of derivative products [30]. It reported that the cellulose hydrolysis is much lower after the increased degree of polymerization of cellulose. The long polymer of cellulose contains many hydrogen bonds between their adjacent free –OH groups which makes it quite stable and decreases the rate of hydrolysis. On the other hand, the shorter chain of cellulose polymers has weak interaction between these –OH groups making it more convenient for enzymatic conversion [31]. Hemicellulose, a heterogeneous biopolymer, contains monosaccharides like xylans, mannans, xyloglucan, glucomannans, and others. Though hemicellulose is amorphous with diminutive physical strength as compared with cellulose, it is more susceptible to break into monomeric units because of the lower polymerism and a reduced amount of crystallinity in its structure. Hemicellulose is more readily hydrolyzed by treatment with dilute acids/bases and also hemicellulase enzymes [32]. Lignin is the second most largely available biomaterial on earth. It is composed of amorphous heteropolymer of phenylpropanoid units (sinapyl alcohol *p*-coumaryl and coniferyl). These polymeric units are interlinked with one other via chemical bonds, for instance, carbon–carbon and aryl ether linkages. It binds cellulose to hemicellulose in the plant cell wall; it acts as a barrier by limiting the polysaccharide accessibility [29].

3 Pretreatment

Pretreatment is one of the important steps for cost-efficient LCB conversion to bioethanol; for over a decade, a wide collection of pretreatment procedures is used on lignocellulosic biomass to produce biofuel with remunerative efficiency [50]. The different pretreatment yields on a variety of biomass are shown in Table 2. The pretreatment process is carried out to reduce the recalcitrance nature of lignin. These include physical, chemical, biological, and their combination pretreatment shown in Fig. 1 in detail [51].

3.1 Physical pretreatment

The physical pretreatment includes mechanical operation such as milling and extrusion, temperature pretreatments such as freezing and hot water, and wave-based pretreatment such as microwave and ultrasound to increase the accessibility of hydrolysable polymer present in lignocellulosic biomass. Microwave and ultrasound pretreatment and mechanical pretreatment followed by hydrolysis of lignocellulosic biomass are the common methods. The mechanical pretreatment method includes milling, chipping, and extrusion methods to improve the bioconversion activity, densification of particles, distribution, and accessibility of enzymatic and comprehensive lignocellulosic biomass transformation into biofuels without formation of any toxic products [67]. Ultrasound pretreatment is applied as a green technology for effective decomposition of lignocellulosic materials, and it plays a significant part in the production of value-added chemicals such as ethanol, methanol, and hydrogen gas [68]. However, this method is energy-consuming; hence, the economic value-added product formation is hampered. The microwave pretreatment method has received attention due to its less reaction time, non-contact volumetric heating and less requirement of solvent. It is used in biorefinery

Table 1 Composition of different lignocellulosic biomass

S. No	Lignocellulosic biomass	Lignin (%)	Hemicellulose (%)	Cellulose (%)	Ash (%)	Ref
1	Sugarcane bagasse	10–20	27–38	40–41	9.0	[33, 34]
2	Rice straw	13–18	24	32	14.89	[18, 35]
3	Barley straw	14–15	22–29	31–34	5.9	[36–38]
4	Switchgrass	31–45	20–21	12–18	5.7	[38, 39]
5	Wheat straw	29–35	26–32	16–21	7.7	[38, 40]
6	Eucalyptus	38–45	12–13	25–37	3.35	[41, 42]
7	Nutshell	25–30	25–30	30–40	6.5	[37, 43]
8	Pinewood	25–30	18–29	40–46	0.3	[33, 44]
9	Hardwood	40–55	24–40	18–25	0.4	[37, 38, 45]
10	Softwood	45–50	25–35	25–35	0.19	[46, 47]
11	Corn stover	7–20.2	34–36	34–36	4.8	[33, 48]
12	Poplar	26.2–28.7	58.8–53	15.5–16	0.59	[46, 49]

Table 2 Pretreatment yield on different biomass

S. No	Pretreatment	Type	Biomass	Optimal conditions	Yield	Ref
1	Physical	Milling	Rice straw	120 min, rumen fluid digestion at 2.5% and 10.0%	Ethanol yield of 116.65 mg/g of 2.5% of the digested residue	[52]
		Extrusion	Vine trimming shoots	55 s, 144.3 L CH4/kg volatile solids	40% increase in methanol production	[53]
		Vibrating ball milling	Corn stover	5–120 min, mechanical grinding	31.23% increased ethyl levulinate yield at 160 °C	[54]
2	Microwaves	Domestic microwave	Cauliflower (*Brassica oleracea* var. *botrytis*)	25 min, 350W	Enhance biogas yield	[55]
3	Ultrasounds	24 kHz, 200 W, horn	Grape pomace	6 min, 10 °C, 50% solid: liquid ratio of 1:10	Increased ethanol	[56]
4	Acid	Diluted H_3PO_4	Green landscaping plan	1.8 h, 12.5 g biomass	Approximately 35% increase in cellulose acetate	[57]
5	DES	ChCl-urea	Sago waste	1 g of sago waste and 20 g of DES, 110 °C for 3 h	Increased glucose yield of 5.2 mg/mL biomass	[58]
		ChCl-LA	Bambara groundnut haulm (BGH)	Pretreatment at 100 °C for 1 h	94.8% glucose yield	[59]
6	Alkali	NaOH (0.5–4% w/v)	Wheat straw	1 h, 121 °C	Highest cellulose concentration of 87.2% was obtained Maximum lignin removal	[60]
7	Organosolv	Ethanol and water (60% wt)	Wheat straw	1.2 h, 200 °C and 400 psi	Approximately 90% lignin extraction at 200 °C	[61]
8	Ionic Liquid	1-butyl-3-methylimidazolium acetate	*Pinus rigida*	6–24 h, 100 °C	Fermentable sugar yield of 92.5%	[62]
9	Combined	Ultrasound-alkaline oxidative ($NaOH$-H_2O_2)	Sugarcane bagasse	10 min, solid loading 20%	94.5% of digestibility of cellulosic fraction	[63]
10	Biological	*Irpex lacteus*	Corn stalks	3–72 h, pretreatment for 0–42 days, 2 g of cornstalks	82% hydrolysis	[64]
		Schizophyllum commune	Paddy straw	7–10 days, 28 °C	50–60% delignification	[65]
		Ceriporiopsis subvermispora	Wheat straw	40 days, 30 °C	34.7% delignification, 23.4% and 66.0% of ethanol yield	[66]

applications as a thermal process [69]. Another energy consuming pretreatment process includes microwaves, which are a form of electromagnetic radiation with frequencies ranging from 0.3 to 300 GHz and wavelengths between 1 and 1 mm. The microwave energy causes rapid heating, particularly in regions with high moisture content which disrupts the biomass structure. The disruption of biomass makes the cellulose more accessible to enzymatic hydrolysis [70].

3.2 Physicochemical pretreatment

The physicochemical pretreatment is a combination of physiological and chemical pretreatment methods. The most studied methods are steam explosion, AFEX and LHW (liquid hot water) pretreatment. The steam explosion is one the most common pretreatment methods used as a physicochemical technique to break the lignocellulosic biomass structure. In this pretreatment process, the materials are exposed to very high pressure and temperature(170–210 °C) for a short period (seconds to several minutes) and then swiftly depressurize the system [71]. It leads to the disruption of the lignocellulosic fibers [72]. However, in LHW pretreatment, the water is used at a very high temperature with high pressure [73]. By this method, the lignocellulosic matrix can be fragmented with a high degree of disrupted polymerization. The temperature ranges from 160 to 240 °C and the time duration range from a few minutes (min) to hours (h), significantly affecting the type of sugar formation and yield of sugar [51]. AFEX is another physicochemical pretreatment technique where the lignocellulosic feedstock is treated with liquefied anhydrous ammonia under extremely high pressure with adequate temperature and then it is quickly depressurized [74]. The moderate temperatures for AFEX are about 60

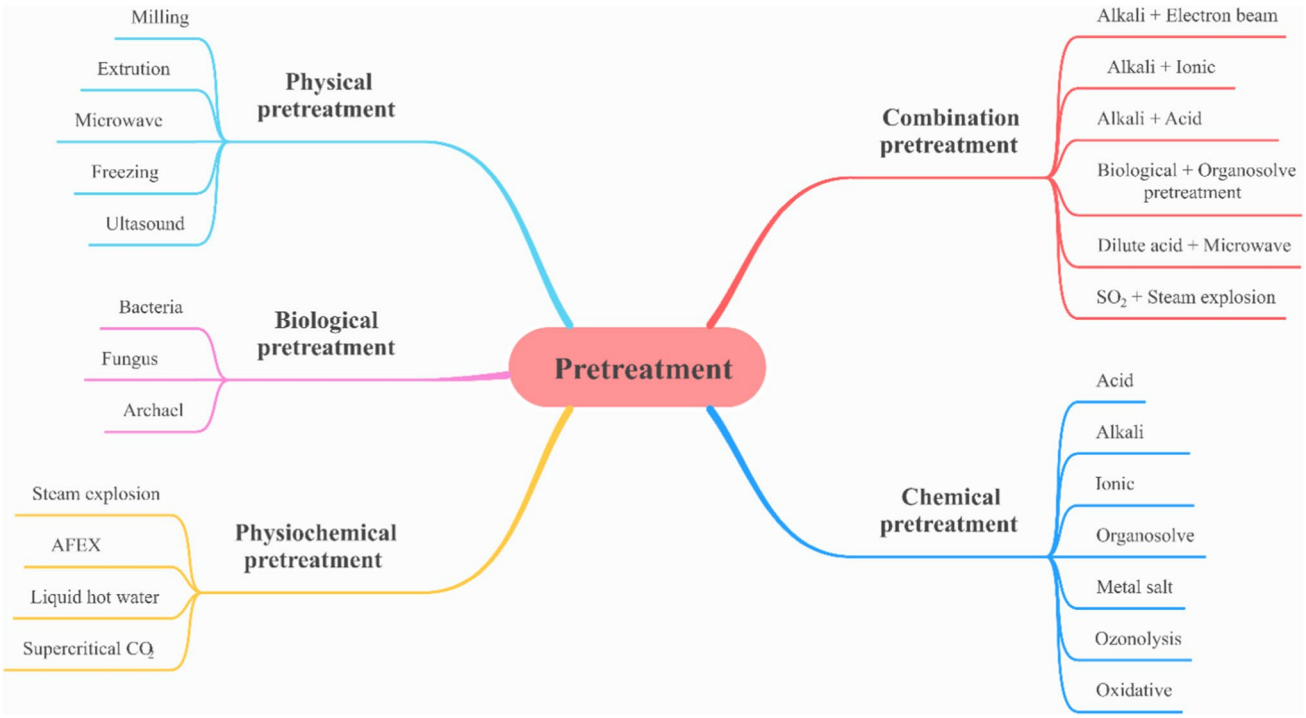


Fig. 1 Methods used for the pretreatment of lignocellulosic feedstock for production of bioethanol [10]

to 100 °C which is considerably lower as compared with the steam explosion process that will overall decrease the cost associated with the process and the amount of energy, which means less energy input and inclusive costs of the process. The average time can be changed from low to moderate, i.e., 5–10 min to 30 min depending upon the degree of saturation required for biomass type [51]. In all the physicochemical methods, the lignocellulosic fiber disruption increases the cellulose accessibility to enzymes during the hydrolysis process. The size of the particle is important for the effectiveness of the process.

3.3 Biological pretreatment

The biological pretreatment method is an environmentally friendly process to remove lignin from lignocellulose. This pretreatment process provides some advantages like low energy input, chemical less process, environment friendly, and economic nature. White rot fungi are most studied among all microorganisms used for biological pre-treatment. The effect of this pretreatment on LCB by using microorganisms including fungus, bacteria, and archaea were estimated on basis of structural and quantitative changes occur in the component of pretreated biomass and vulnerability to enzymatic hydrolysis [75]. The enzymatic pretreatment method is another way of biological pretreatment process, where enzymes (either produced or procured) are loaded on the feedstock for disruption of biomass structure. Cellulase is the most common enzyme used for this process. But the most common disadvantage of biological pretreatment is the slow rate of hydrolysis thus, less conversion of monomers [76].

3.4 Chemical pretreatment

Chemical pretreatment methods such as acid, alkali, ionic, and organosolv are the traditional pretreatment methods, while DES pretreatment is the new class of ILs, used due to their effective bio-degradative nature [77]. The chemicals which are used for such treatment process include alkali such as sodium hydroxide, potassium hydroxide, calcium hydroxide, and ammonium hydroxide [78, 79]; acid such as formic acid, maleic acid, and oxalic acid, sulfuric acid, phosphoric acid, nitric acid, and hydrochloric acid [80, 81]; and organic and inorganic ILs [82, 83]. The chemical pretreatment method increases the surface area of biomass by swelling, reduction in crystallinity, degree of polymerization, damage of connection between the lignin and other polymers, and also lignin breakdown [84]. When the lignin content is low in the biomass, the chemical pretreatment works better. The disadvantages of this method are low sugar and inhibitor formation and the requirement of high temperature (Table 3) [85].

Table 3 Advantages and disadvantages of various pretreatment processes

S. No	Pretreatment methods	Main effect	Advantage	Disadvantage	Ref
1	Mechanical	Reduction in crystallinity of cellulose	No inhibitor formation	Consumption of high power and energy	[86]
2	Concentrated acid	Hydrolyses both cellulose and hemicelluloses	High glucose generation Due to moderate temperature, operational cost is reduced Few degradation product formations No requirement of enzymes	Recovery of acid is mandatory Equipment corrosion Formation of inhibitory compounds	[87]
3	Alkali acid	Hydrolyses the hemicelluloses, makes cellulose more manageable to enzymatic treatment, alters the structure of lignin	Low inhibitors formation Low corrosion problems than concentrated acid	Due to high temperatures degradation product form Low concentration of sugar formation	[88]
4	Ozonolysis	Lignin content is reduced	No inhibitor formation Mild operational condition	Expensive ozone A large amount of ozone is needed	[89]
5	Organosolv	Hydrolyses, hemicelluloses and lignin	Pure recovery of lignin High digestibility	High costs of Solvents requirement Need to be drained and recycled	[90]
6	Steam explosion	Solubilization	It's cost-effective Higher yield of glucose and hemicelluloses	Inhibitory compound formation Partial deprivation of hemicelluloses	[91]
7	Ammonia fiber explosion (AFEX)	Increase available surface area Lignin and hemicellulose are removed to an extent	Low inhibitor generation	Not good for the biomass with more lignin	[91]
8	Liquid hot water	In LCB it substantially lowers the cell wall recalcitrance Enhance polysaccharide scarification	High recovery rates for pentoses Generates low number of inhibitors	Degradation of monosaccharide sugars	[91]
9	Supercritical CO_2	Hydrolyze hemicellulose and increase substrate accessibility for enzymatic hydrolysis	Increases accessible surface area Does not produce inhibitory compounds Nonflammable and low toxicity	Does not alter lignin or hemicelluloses High cost	[89]
10	Biological	Degrades lignin and hemicelluloses	Consumption of Low energy	Slow hydrolysis rate	[92]
11	DES	Removes lignin and hemicelluloses Drastically reduce biomass recalcitrance	Ease of preparation, easy availability, and economically feasible Low vapor pressure, wide liquid range, non-inflammability	Toxicity, on biodegradability, complex synthesis require purification, and high cost of the starting material	[93]

3.5 DES pretreatment

DES (deep eutectic solvent) was first presented by Abbott and his colleagues that it is a eutectic mixture that contains hydrogen bond donors (HBD) (Fig. 2), e.g., amines, carboxylic acids, amides, alcohols, and hydrogen bond acceptors like quaternary ammonium salts that have the ability to alter the physical structure of the biomass with minimum consumption of energy during the process. It is thought that the binding of quaternary salt with hydrogen bond donors, in DES, will reduce the electrostatic force of cation–anion-forming hydrogen bonds, resulting in a substantial decrease in the freezing temperature of the mixture. As per the literature, the most common DES pretreatment is a combination of choline chloride/lactic acid. It is reported that a mixture of these two solvents, utilized in DES pretreatment, can attain exceptional results biomass pretreatment [94, 95]. The bond between the lignin-carbohydrate complexes (ether bond) can be broken by the solvent, generating exposed surface and holes for enzymes which help in degrading the cellulose [81–83]. DES reagents are biocompatible and exhibit very low toxicity to the environment and are barely volatile. It can also be reused again for the next pretreatment process [95]. DES contains the ideal solvent properties of ILs that include low volatility, non-toxicity, biocompatibility, and wide liquid range [18]. In recent past, the toxicity of DES is much discussed [96]. Due to their potential contribution towards environmental pollution. However, many researchers have also shown the biodegradability of DES solvents which ranges between 60 and 90% within 14 days of microbial

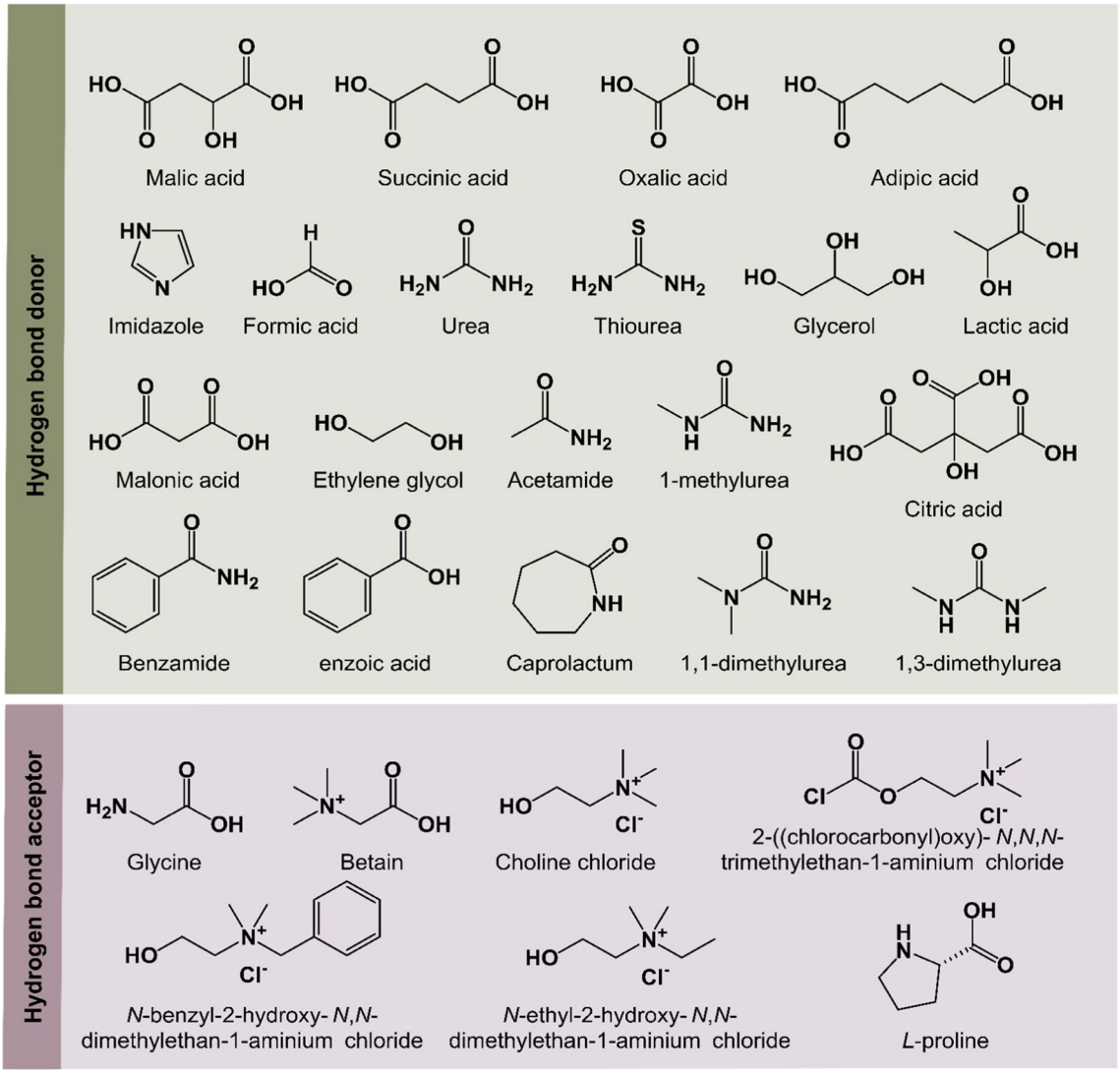


Fig. 2 Structures of various hydrogen bond donors and hydrogen bond acceptors for DES [103]

incubation [96–98]. The work conducted by the group of researchers on the toxicity of DES reported that the toxicity is not because of its chemical composition but due to media-induced acidity. The cell can tolerate a concentration of 300 to 450 mM of DES [99]. Moreover, the toxicity determination by different methods such as antibiogram, well diffusion method, and microtox, and MTT assay is not much accurate because of the turbid solution of the DES solvents and low diffusion ratio [96]. DES is cost-effective and environment friendly, making it a viable alternative for the pretreatment of feedstock. Additionally, the recyclable nature of the solvent with a greater recovery percentage makes it more economic as compared with any other pretreatment procedure [100]. Kumar et al. [101] performed an experiment with ChCl/lactic acid and extracted 68 mg/g of biomass lignin. The recycled DES solvent was capable of extracting lignin without considerable loss. Additionally, according to Bradic et al. [102], DES containing lactic acid demonstrated satisfactory extraction efficacy after two cycles.

4 Deep eutectic synthesis

The molar ratio of HBD & HBA is an important parameter that affects the DES preparation. The different structures and concentrations of HBD/HBA affect the physicochemical properties of DES like surface tension, viscosity, melting point, freezing point, acidity or alkalinity, conductivity, polarity, refractive index, density, boiling point, solubility, and so on. The biomass pretreatment with ionic liquid is a costly due to the intricate synthesis process. Therefore, DES is observed as a cheaper substitute for ionic liquids. Smith

et al. [23] describe a general formula to demonstrate the DES synthesis process [104, 105].

$$Cat + X \rightarrow zY$$

whereas cat is the cationic group; X is the Lewis base; y is the Lewis or Brønsted acid with z number of molecules interacting with anion [106].

Since there are many possibilities of formation of various types of DES which can exhibit unique physicochemical properties between 50 and 100 °C, these must be classified for their uniformity and easy identification [107].

4.1 Classification of deep eutectic solvents

DES can be classified into four different categories based on their composition as represented in Fig. 3.

4.2 Type-I DES [composed solely of ionic species]

The composition of type-I DES is a mixture of quaternary ammonium salts (choline chloride) and non-hydrated metal halides ($AlCl_3$, AgCl, and $CuCl_3$), but due to excessive melting points, it has limited application in biomass bioprocessing. These solvents are less toxicity can be easily synthesized from readily available components and biodegradability; these solvents are frequently seen as environmentally beneficial alternatives to conventional organic solvents. They are used in several disciplines, such as material science, chemical synthesis, and extraction techniques [23]. Rehmann et al. [108] reported the effective pretreatment of corncob with ChCl/imidazole for efficient enzymatic saccharification.

4.3 Type-II DES [composed of water]

The type-II DES is mainly composed of metal halide hydrates (like $AlCl_3{\cdot}6H_2O$) combined with quaternary ammonium salts. Type-II DES have a minimal melting point at a particular composition, known as their eutectic point. Due to these characteristics, it remains in a liquid state at low temperatures [23]. It is viable for various industrial processes due to the lesser cost of halide salts easy availability [109]. The DES of combination of zinc chloride (HBA) and ethylene glycol (HBD) are reported to have higher degree of extraction of heavy metal and organic contaminants. The solvent has low environmental impact compared with other conventional solvents [110].

4.4 Type-III DES [salts and molecular compounds]

It is a combination of quaternary ammonium salts and hydrogen bond donors such as amide alcohols and carboxylic acids. It is the most commonly used eutectic mixture for biomass processing because of the easy preparation, affordable materials, lower viscosity, and excessive biodegradability [59]. Though one of the components of these DES is a metal salt, they have qualities similar to other types. These solvents exhibit lower melting points than their individual components, resulting in a eutectic mixture. This solvent has drawn the interest of researchers due to its excellent conductivity, adjustable qualities, and eco-friendliness, and it exhibits numerous applications in materials science, electrochemistry, and catalysis [111]. Among various combinations of DES, the ChCl/Lactic acid is very effective in biomass delignification at 1:5 to 1:10 molar ratios. Additionally, ChCl in combination with alcohol exhibit high lignin extraction [112].

4.5 Type-IV DES [mixtures of two molecular compounds]

Type-IV DES system is a eutectic mixture of metal salts and HBD-like amide and polyols. Metal salts usually ionize in

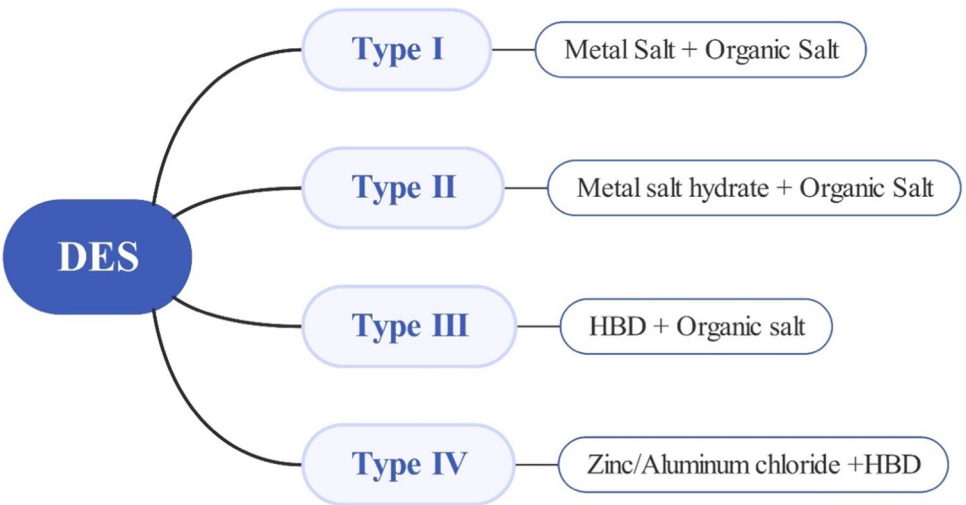


Fig. 3 Classification of DES

non-aqueous solvents, but $ZnCl_2$ is shown to form a eutectic solvent with urea, acetamide, ethylene glycol, and 1,6-hexanediol [23]. Zinc chloride and donor molecules such as acetamide, urea, or ethylene glycol have significantly different viscosity, conductivity, density, phase behavior, and surface tension as compared with the ionic liquids that incorporate quaternary ammonium salts [113].

5 Physicochemical properties of DES

The deep eutectic solvent containing HBA and HBD has unsymmetrical ions. It has low lattice energy and a very low melting temperature [23]. Various kinds of DES and their special chemical composition provide them with different physicochemical properties which make them useful for different industrial processes. The industrial application of DES makes it a more attractive solvent, and it warrants an urgent need to study its physicochemical properties such as freezing point, density, viscosity, surface tension, and conductivity as discussed below:

5.1 Freezing point

One of the main properties of DES is its freezing point compared with its constituents. The freezing temperature of eutectic solvent tends to decrease due to the charge delocalization as a result of hydrogen bonding between halide ions and HBD. The freezing point of the DES was also found to be affected by the internal energy of DES solvent during liquid formation [23]. The reduction in freezing point is also accompanied by the disruption of crystalline structure. The DES solvent available at present shows freezing points below 150 °C, but DES solvent with a freezing point less than 50 °C has increased the interest for various industrial applications as a solvent [114]. Figure 4 shows the phase transition of HBA and HBD.

The increase in the hydrogen bond strength due to eutectic solvent composition is counterbalanced by other cohesive interactions. The factors that affect the melting point of DES are lattice energy, entropy change during DES formation, and the anion and HBD interaction. According to Abbott et al. during the eutectic solvent formation of ChCl and urea at 1:2 molar ratios, the larger melting point depression was observed due to the charge transfer via H-bond between the urea and the chloride ion [115].

5.2 Density

The density is considered as a thermophysical property, which is studied for different DES solvents. The most common DES solvent used today has higher density than water [116]. For example, type-IV DESs often have densities > 1.3 g/cm^3, and the mixture of metallic salt-based solvents such as $ZnCl_2$:ethylene glycol and $ZnCl_2$:urea has a high density ranging from 1.3 to 1.6 g/cm^3 [107]. Hole theory is used to depict the density properties of DES solvent; the density behavior is very much affected by the packing and arrangement of the molecules present in DESs [117]. According to this theory, a combination of metal salts and HBD shows the reduction in average hole radius which causes the decrease in density of DES solvent as compared with pure urea [116]. Moreover, the DES solvent density also tends to decrease with an increase in the alkyl chain length of solvent components and an increase in the relative salt ratio to HBD. Besides this, the increase in water content of the DES system also results in a decrease in density [86].

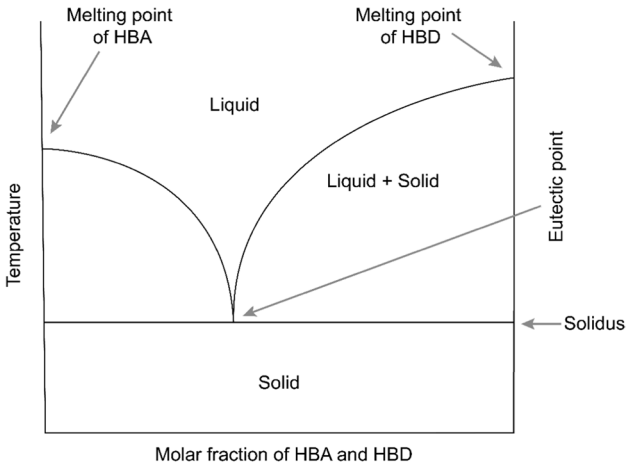


Fig. 4 The phase diagram representing eutectic mixture of two components

5.3 Viscosity

The DES viscosity depends on the intermolecular interaction which is influenced by many factors, for example, the type of HBA, HBD, temperature, molar ratio, and the water content present in the solvent. The DES's viscosity is due to the fairly large ion size and free space between the molecules in the ionic system [23]. However, studies in the past show the decrease in the DES viscosity is associated with the increase in temperature and amount of quaternary ammonium salts [88, 89]. On macro-scale industrial applications, the low viscosity of DESs is required when considering the economic and technical aspects of downstream processing. Therefore, the hole theory can be used to design the low viscosity DES [118]. Despite the poor conductivity of the DES solvent, it is in great demand for the replacement of organic solvent due to its high stability and biodegradable nature [55, 111].

5.4 Surface tension

The DES surface tension is dependent on the dominant intermolecular forces and the variety of cations. The hydroxyl group of cations shows higher surface tension because of its ability to bind hydrogen [91, 104]. Therefore, the glucose-based DES solvent has higher surface tension than carboxylic acid-based DESs [119]. The surface tension of the DESs solvents decreases with an increase in temperature because of the reduction in intermolecular forces. One of the remarkable properties of DES is to adjust its properties by the careful selection of HBD and HBA and by varying its molar ratio based on its application [90, 91].

5.5 Conductivity

The conductivity is correlated with viscosity; the higher the viscosity, the lower the conductivity. The most common DES has very low conductivity at room temperature [120], which can cause problems for particular electrochemical applications. The low viscosity DES like ethylene glycol or imidazole with ChCl exhibits enormous conductivities [114]. The hole theory was applied by Abbott et al. proposed the conductivity of the DES system; however, some good results were obtained in a few cases like DES containing ChCl and ethylene glycol in 1:2 ratios, and poor observations were reported in other cases like DES of ChCl and glycerol with 1:2 ratios. According to this, the mobility of ions and the conductivity can be determined not only by the presence of suitable holes but also by the type and strength of ion-HBD interaction. Therefore, it should be noted that ions migrate while tangled with corresponding HBD, and the size of HBD will determine the hydrodynamic radius of migrating species, which can be used to determine its transport properties [116]. The conductivity of DES is increased by increasing the concentration of salt (IL), but it's not necessarily true for every DES. However, the conductivity variation with salt depends on both the salt type and the HBD [107].

6 Categories of hydrogen bond donor (HBD)

The overall performance of DESs varies with the different categories HBD's and HBAs as well as types of compounds within the same category.

6.1 Carbohydrate-based DES

DES-based on carbohydrates is composed of sugar compounds as hydrogen bond donors like xylitol glucose and fructose. The pH of these DES solvent is usually neutral when it mixed with choline chloride, common HBA. Apart from its mild nature, ChCl:fructose DES can dissolve up to 0.65% weight of powdered rice straw, i.e., 6.5 mg/g of DES [121]. On comparing it with aldoses, DES based on fructose confirmed high dissolution ability with less viscosity. However, carbohydrate-based DESs has the ability to dissolve the biomass, but it has very low dissolution ability as compared with the other DES system. Thus, these sugar-based DES are not generally used in lignocellulose bioprocessing; they have application in other fields like food or pharmaceutical processing [122].

6.2 Hydroxyl-containing DES

DES solvent is synthesized by using HBD with hydroxyl groups and is most widely used, including polyalcohol and carbohydrate-based hydrogen bond donors. These kinds of solvents generally show biocompatibility with microbes and the enzymes at neutral pH when mixed with the nontoxic hydrogen bond acceptor (e.g., betaine, ChCl). Hydroxyl-containing HBD are neutral in nature, and many works show low lignin dissolution and fractionation with polyalcohols or carbohydrate-based DESs, even when it's under severe pretreatment conditions. DES pretreatment containing hydroxyl group as HBD is good for grass pretreatment, because of its weaker covalent bond interaction hydroxyl group plays a positive role during lignin extraction [86, 88]. The number of hydroxyl groups in the hydrogen bond donor is inversely associated with lignin removal. For example, ChCl/glycerol exhibited worse lignin extractability than ChCl/ethylene glycol [123]. For instance, the efficiency of pretreatment and ability of lignin removal with polyalcohol-based DES system is reduced in the order of ethylene glycol > glycerol > xylitol, when lactic acid is used as HBA [124]. Ethylene glycol has two hydroxyl groups, and it can interact more strongly with lactic acid which could result in a more effective DES system for the removal and dissolving of lignin. Glycerol contains three hydroxyl groups; it is slightly less effective and capable in removing lignin than ethylene glycol. The reason for the marginally lower DES performance could be attributed to variations in molecule structure or interactions with lactic acid. Xylitol, a polyalcohol with five hydroxyl groups, seems to be the least effective. Despite containing more hydroxyl groups than glycerol, xylitol may interact with lactic acid differently on a molecular level, which could result in less efficient lignin removal. Therefore, throughout the DESs pretreatment, a low amount of delignification rate was related to a more hydroxyl group in the hydrogen bond donors [125].

6.3 Acid-based DES

Acid-based DES system is applied in the bioprocessing of biomass containing organic acid as solvent. Mixtures of

compatible hydrogen bond acceptor and acid-based DESs achieve improved results in the pretreatment of biomass. According to the literature, the removal of xylan and lignin were increased to 30% when treated with acid-based DES in place of organic acid [126]. The acid-based performs well in xylan hydrolysis when pretreated at 100 °C. However, in DES solvent, the lignin and xylan were solubilized, but the main portion of cellulose was retained in DES-pretreated solid biomass. It is reported that 86% and 98% of glucose were produced by formic acid in the pretreated oil palm and herbal residues, respectively [95, 107], which can affect the downstream processing efficiency. The DES fractionation efficiency is greatly affected by the different functional groups at varying amounts; acid hydrogen bond donors containing multiple carboxyl groups (diacid or triadic) can decrease the efficiency of fractionation as compared to mono acid hydrogen bond donors. Apart from the key functional group in acids, i.e., carboxyl group, other chemical groups that are found in acid also influence DES performance, like the presence of electron-donating and electron-withdrawing groups [95, 109].

6.4 Alkaline DES

In alkaline-based DES pretreatment, alkaline solvents are usually used for the extraction of lignin from lignocellulosic biomass because of the alkaline soluble nature of lignin. For instance, when the biomass containing holocellulose, acid-insoluble lignin, and α-cellulose was pretreated with the mixture of ChCl/urea, the lignin shows excessive solubility as compared with other chemical mixtures. The alkaline pretreatment will disrupt the lignin as well as lignin-carbohydrate compounds by breaking ether bonds and leading to lignin depolymerization repolymerization[127]. Likewise, basic IL pretreatment cleaves ether bonds and leads to demethoxylation, dehydration, and repolymerization primarily through β-β′and β-5′ linkages [128]. The authors also reported 51% lignin and 36% hemicellulose with DES solvent K_2CO_3 and glycerol [129].

7 Categories of HBAs

The high productivity of DES delignification is recognized by the collaborative effect of HBDs and HBAs. HBA includes phosphonium salts, sulfonium salts, and ammonium salts. HBD includes amide alcohols and carboxylic acids [103]. The most widely used HBA in DES pretreatment is ChCl because of its biodegradable nature, non-toxicity, and low cost. HBA plays an important role in the performance of DES pretreatment [130, 131]. The process of lignin removal with ChCl/formic acid was evaluated with a pure solution of formic acid, and the result obtained showed approximately 40.7% of lignin removal while using ChCl/formic acid and merely 9.0% removal from herbal residues on treatment with pure formic acid [121]. Therefore, DES exhibits enhanced lignin fractionation and xylan removal than the corresponding acids alone. In a study, DES pretreatment with hydroxyl HBA (ChCl) resulted in 82.70% glucan retention, slightly lower than pretreatment with HBA containing chloro allyl trimethyl ammonium chloride (ATMAC) or carbon–carbon double bond (CCC) groups, which retained around 86% of glucan [132]. DES containing carboxylic acid (lactic acid-LA) with ATMAC as a HBA contains C–C double bond. It is observed that the solvent removes nearly 70.24% of Xylan as compared with the pretreatment with ChCl/LA which removes approximately 78%. Therefore, a poor delignification was observed with the double bond CCC-LA [125].

The other categories of HBA includes polyol alcohols, such as EG, glycerol (Gly), butanediol, 4-hydroxybenzyl alcohol, pyrocatechol, xylitol, 1,2-propanediol, and 1,3-propanediol, which represent a category of 1,3-propanediol. Even though it requires more severe conditions than acid-based HBDs, polyol HBD exhibits a various lignin removal rate depending on parameters like temperature, reaction duration, and the arrangement of molecular and functional groups [133]. The current research shows that the extremely electronegative nature of Cl^- in ChCl tends to form hydrogen bonds with hydroxyl group present in lignin, which helps in its removal. Other researchers found that the halide counter ion in hydrogen bond acceptor had enabled the breaking of β-O-4′ bond and stops lignin self-condensation [134]. However, betaine-based DES pretreatment containing the same halogen ion dissolves 12% of lignin, while DES pretreatment with ChCl-based dissolved 100% of lignin. This suggests that the productivity of lignin removal is greatly affected by the coordinating ability of cations in the hydrogen bond acceptor [134]. However, the DES solvent with different ratios of HBD and HBA at optimal conditions is also effective in the solubility of the cellulose and lignin (Table 4). DES of malic acid and proline 1:3 molar ratio exhibits the highest lignin solubility and lowest cellulose solubility [122]. ChCl and formic acid DES molar ratio 2:1 is the second highest in the lignin solubility and < 1% of the solubility of hemicellulose and lignin. LA and betaine DES at molar ratio 2:1 (Table 4) are also able to solubilize lignin [122]. Various DES solvents along with their solubility and molar ratio are listed in Table 4.

8 Biomass pretreatment with DES solvent

Biomass pretreatment is important to achieve high saccharification in turn yields greater ethanol production (Fig. 5). Several other pretreatments are not effective in the removal of lignin and in harsh conditions like concentrated acid

Table 4 Solubility of cellulose and lignin in different DESs

S. No	Hydrogen bond donor	Hydrogen bond acceptor	Molar ratio	Temp. (°C)	Lignin (%)	Cellulose (%)	Hemicellulose (%)	Ref
1	Lactic acid	Choline chloride	2:1	60	7.56	0	-	[122]
2	Lactic acid	Alanine	9:1	60	8.47	0.00	-	[122]
3	Lactic acid	Proline	3:3:1	60	9	< 1	< 1	[26]
4	Lactic acid	Glycine	9:1	60	8.77	0.00	-	[122]
5	Lactic acid	Betaine	2:1	60	9	< 1	< 1	[26]
6	Urea	Choline chloride	2:1	110	-	< 0.2		[97]
7	Acetamide	Choline chloride	2:1	110	-	0.22	-	[135]
8	Formic acid	Choline chloride	2:1	60	14	< 1	< 1	[26]
9	Imidazole	Choline chloride	7:3	110	-	2.48	-	[135]
10	Ammonium thiocyanate	Choline chloride	1:1	110	-	0.85	-	[26]
11	Malic acid	Proline	1:3	60	14.90	5.90	-	[122]
12	Zinc chloride	Choline chloride	2:1	110	-	< 0.2	-	[118]
13	Caprolactam	Choline chloride	1:1	110	-	0.16	-	[136]
14	Oxalic acid	Allyl triethyl ammonium chloride	1:1	110	-	6.48	–	[136]
15	Oxalic acid dihydrate	Choline chloride	1:1	60	3.62	2.5	-	[122]

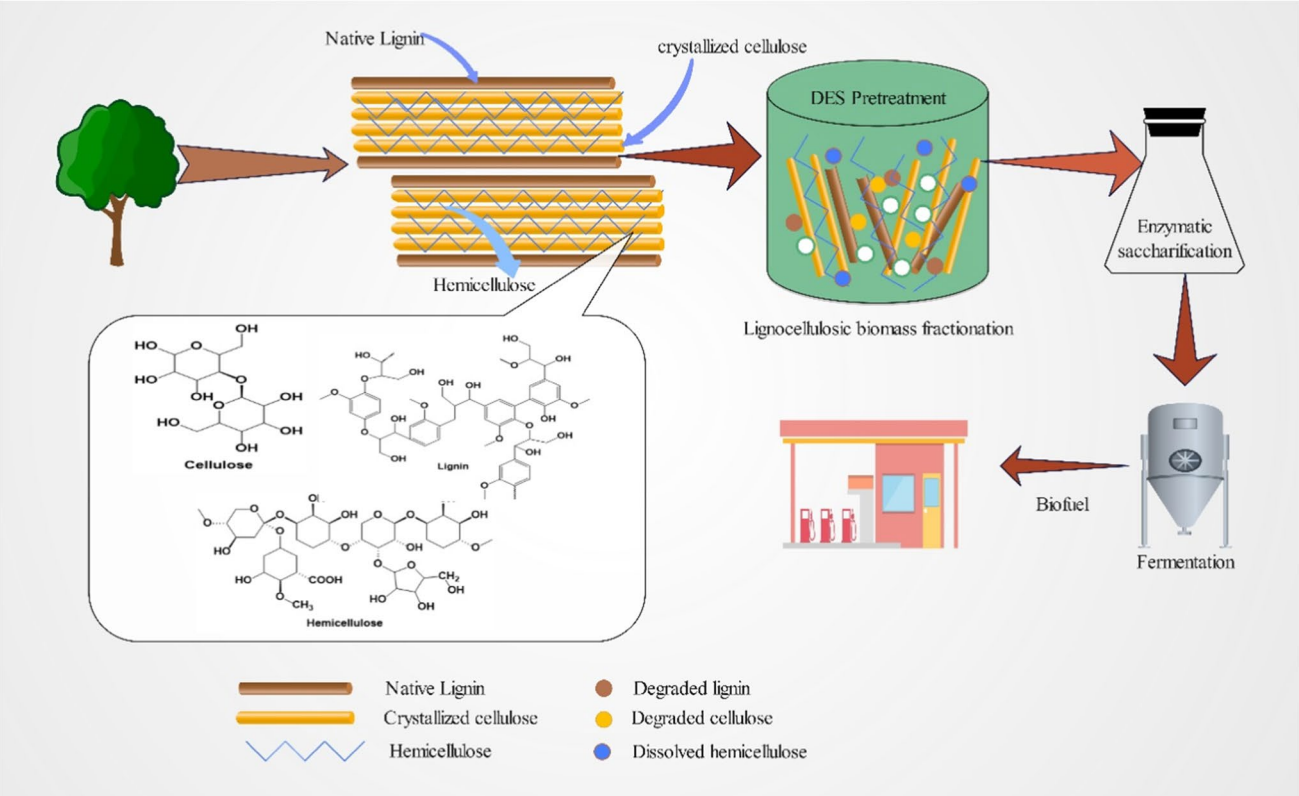


Fig. 5 Lignocellulosic biomass fractionation with the DES solvent and fermentation for alcohol production

pretreatment cause the degradation of hemicellulose and inhibitor formation [137]. However, lignin solubility in DES was proved to be more effective and hence attracted the attention of researchers to explore as new alternative for the pretreatment of biomass. Recent studies on DES with different ratios of HBD and HBA suggested that acidic DES

solvent has shown more than > 90% delignification of lignin from a variety of different lignocellulosic feedstocks, for example, rice straw, wheat straw, corn cob, and poplar [128]. However, pretreatment with ChCl:glycerol alone is not very effective unless it is treated with a hydrothermal process to reduce the recalcitrance before DES pretreatment [138] and develop a new class of DES with lignin-derived phenols as hydrogen bond donor and 4-hydroxybenzyl alcohol as hydrogen bond acceptor for delignification of switchgrass [139]. In contrast, on the treatment of biomass with dilute sulfuric acid, approximately 80% of the most common hemicellulase can be hydrolyzed into soluble sugars, for example, xylose, galactose, and arabinose mannose, at 120 °C temperature and pH from ~ 1.5 to 2 for 30 min [140]. Thus, pretreatment efficiency and degree of delignification of DES are greatly depended on the recalcitrance nature of the biomass and the pretreatment conditions. The various pretreatment conditions like temperature, the ratio of solid to liquid, and time can affect the performance of the solvent [141]. Temperature plays a vital role in DES preparation and performance due to their phase transition from solid to liquid (Table 5). According to Hou et al., freezing and melting points of DES greatly influenced by factors such as ionic symmetry, alkane chain length, and hydrogen bonding capacity [142]. DESs with freezing temperatures lower than 50 °C are the ideal choice for biomass processing [143]. DESs have been extensively applied for the delignification, cellulose extraction, and nanofibrillation because of its selective solubilization. The ChCl-lactic acid, betaine-lactic acid, or certain amino acid-based DES solvents are helpful in the dissolution of lignin and have a very low or insignificant solubility in cellulose in terms of biomass separation applications. Thus, it has been used to selectively delignify wheat straw [122]. It is an advantageous and less hazardous substitute for ILs from an economic standpoint [144].

9 Biocompatibility of DES with enzyme and microorganism

This solvent has attracted researchers due to its biocompatibility with biomolecules such as protein, nucleic acid, enzyme, and microorganisms. The impact of DES on cellulase was assessed where cellulase enzyme was isolated from *Aspergillus* species with chloride-based DES solvent with glycerol or malonic acid or ethylene glycol as the hydrogen bond donors in 1:2 ratios for 48 h. After 48 h, in control, the observed result showed a reduction of 10% in filter paper activity (with no DES solvent), while a loss of 60% was observed in filter paper activity by using DES solvent comprising malonic acid within 24 h [155]. According to Hayyan et al. and co-worker, DES solvent is used to check the compatibility with the microbes [156]. Therefore, four bacterial strains, viz., *Bacillus subtilis*, *Escherichia coli*, *Staphylococcus aureus*, and *Pseudomonas aeruginosa*, were taken to perform the filter paper assay (24 h) to see the toxicity that is caused by choline chloride DES solvent with glycerol, ethylene glycol, triethylene glycol, and urea as HBDs [138].

Table 5 Applications of DESs with different biomass

S. No	Biomass	DES	Ratio	Time (h)	Temp (°C)	Extraction	Further conversion	Ref
	Rice straw	ChCl:formic acid:acetic acid	1:1:1	2	140	Glucose extraction	Butanol fermentation	[145]
	Wheat straw	TABAC:lactic acid (1:9) (2,3-Dihydroxypropyl-1-triethylammonium chloride)	1:9	10	100	Higher extraction of lignin and complete fractionation of biomass	-	[146]
	Switchgrass	ChCl:ethylene glycol	1:2	0.5–0.75	130	Lignocellulose fractionation	Furfural, 2,3-butanediol	[147]
	Industrial zylose residue	ChCl:lactic acid	1:2	2	120	Extraction of Cellulose and lignin	Enzymatic hydrolysis	[148]
	Crustacean waste, i.e., shrimp shell	ChCl:malonic acid	1:2	2	80	Chitin extraction	Chitin films	[149]
	Corn stover	ChCl:formic acid	1:1	2	130	Pretreatment	Butanol fermentation	[150]
	Rice flour	ChCl:tartaric acid	1:2	1	60	-	Removal of cadmium	[151]
	Tea leaves	ChCl:1,4-butanediol ChCl: malonic acid ChCl: acetamine	1:3,	0.006	-	Alkaloids, flavonoids, and catechins	-	[152]
	wool	ChCl: Oxalic acid	1:2	2	110–120	Keratin	-	[153]
	Oil Palm Fronds	ChCl: Urea	1:2	4	120	High percentage of hemicellulose	Xylan recovery	[154]

In contrast, in other studies, the researcher performed the same assay with a similar condition on the same bacteria with phosphonium-based DES (methyl triphenylphosphonium bromide) having HBDs as glycerol, ethylene glycol, and triethylene glycol; HBDs seems to inhibit the same four bacteria [156].

10 Industrial application of DES

DESs exhibit significant potential in wide range of application. The first application of DES which sparked the interest of researchers was in the field of metal processing and media synthesis. The DES solvents of metal and metal salts have high solubility and electrical conductivity making it an ideal solvent in the extraction of metal [23]. Another significant area of application includes biomass processing for production of bioethanol and biochemicals through fermentation procedure [157]. DES's sustainable and versatile nature has made it a promising extraction solvent of complex food samples. Different eutectic mixtures of some organic salts and alcohols, acids, etc. can successfully extract contaminants from food such as pesticides, heavy metals, plastic migrants, and biomolecules [158]. Additionally, DES exhibits potential in the extraction of bioactive compounds from different plants due to its high thermal stability and low flammability [159]. The DES is reported as a green solvent due to its exceptional characteristics, high solubility, and biodegradability; DESs have industrial uses in medication administration, electrochemistry, natural product extraction, and trace metal analysis [160].

11 Environmental impact of DES

The DES is considered more safer as compared with other chemical pretreatment processes. The alkaline and acid treatments are known to yield certain toxic compounds that can cause harmful effects. Hydroxymethyl furfural is formed during the hydrolysis of sugars by acids and acts as an inhibitor of yeast fermentation. HMF is toxic to the microorganisms used in ethanol production, hence reducng the activity and overall yields of ethanol [161] [162]. Apart from HMF, organic acid like acetic is also produced. Though it is less toxic than HMF but higher quantity reduces the efficiency of fermentation [163]. The DES, on the other hand, is considered as green solvent due to lesser-known toxic impact on health and environment. However, some studies suggest that the DES solvent used in the pretreatment process can be harmful to the aquatic organism and disrupt the ecosystem. It is due to some compositional elements which cannot degrade biologically. Hence, accumulation of such components in the soil and water bodies may cause threat to the wildlife and humans [163].

In one of the major concerns regarding water body pollution, it was investigated that electroplating application of DES leads to electrolysis which contains heavy metals. These heavy metals can be treated using ion exchange, making it more safer as compared with ILs [164].

12 Computer simulation studies of DES

The DES solvent has several limitations that hinder its application despite its promising properties. One of the major drawbacks is high viscosity and low volatility of DES restricting its application [165]. Additionally, insufficient understanding of the basic interaction of DES and biomass polysaccharides makes it an ideal choice for computer-based simulation to understand the interaction properties [123]. Molecular dynamics (MD) simulation has been utilized to investigate various DESs, such as choline chloride and ascorbic acid, where the results indicate that hydrogen bonding networks are stable and remain unaffected by large temperature variations ranging between 303.15 and 353.15 K [166]. Conventional simulation strategies are also utilized for interaction studies of DES with monosaccharides as well as polysaccharides through radial distribution function leading to the dissolution of glucose and the process of surface deconstruction [167]. These methods are useful in studying the LCB disintegration by identification and prediction by microscopic interaction [168].

Four different DES have been investigated at room temperature, viz., choline-chloride ethylene glycol, choline-chloride oxalic acid, choline-chloride urea, and choline-chloride levulinic acid, to investigate the correlations between the Kamlet–Taft α and β values with several thermodynamic, structural, and energetic parameters, which describe how an atomistic interaction might hydrolyze cellulose. Kamlet–Taft α and β parameters are used to characterize solvent properties. The study revealed that the weaker interaction between DES leads to the stronger interaction of DES and the cellulosic components of the biomass. However, both Kamlet–Taft α and β parameters are important, rather than separately for the development of sustainable solvents [169]. The potential of DES and their application to enhance biomass pretreatment processes through machine learning including principal component analysis, partial least squares, linear regression, optimized gradient boosting, artificial neural networks, and random forests. Machine learning provides the appropriate prediction for the effectiveness of DES pretreatment of lignocellulosic biomass with performance indicators such as R-squared values and root mean square error (RMSE), leading to more efficient and cost-effective biomass processing [170].

13 Conclusion

The need to minimize dependency on fossil fuels by replacing them with economically viable and ecological friendly alternatives has grown over the years. A number of societal, economic, and environmental issues are intended to be addressed by this change. The issues of resource depletion, waste management, and the loss of valuable biowaste that contains energy must all be addressed for sustainable growth. A major challenge in the commercial application of biomass processing to ethanol is the pretreatment process, which requires an economic and green solution. The efficient hydrolysis can be hindered by components such as cellulose crystallinity, degree of polymerization, moisture content, surface area, and lignin content. The pretreatment's primary objective is to extract lignin and disrupt the structure of lignocellulose, which is usually accomplished by thermochemical methods. The process of converting lignocellulose to fermentable sugars, where biomass-to-sugar conversion is expensive but technologically accessible. The use of DES is proved to be a potential pretreatment method for biomass valorization due to its unique physicochemical properties. DES is considered as cost-effective, environment friendly and less inhibitor-producing pretreatment method and thus can be termed as green pretreatment. It can be considered a sustainable alternative for biomass processing.

14 Future prospect of DES

Despite the fact that DES is being applied in various industrial applications, its use in biomass processing is less explored. There is a need to explore the various possibilities of different combinations of DES for biomass pretreatment and its impact. The commercial application of lignocellulosic biomass is the direction of future drifts. Improving the effectiveness of lignocellulose and cogeneration for sustainable biofuels and other products should be the main focus of future research. Future research efforts should primarily focus on lignocellulosic biomass and green chemistry as means of mitigating the current energy crisis and environmental contamination. Therefore, research should be focused on the generation of workable, affordable, and environmentally acceptable clean alternatives for the sustainable utilization of lignocellulosic biomass. This review provides an insight of use of DES solvents for use in the pretreatment of lignocellulosic feedstock. There are a majority of different combinations of hydrogen bond donor and hydrogen bond acceptor possible which can be utilized for various applications in the biorefinery process as well. Further investigation is needed on its application and scale-up process using the DES system to reveal its full potential as "Green" solvent and also to add new possibilities in the biorefinery industry.

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Data availability The data that support the findings of the study are available in the manuscript**.**

Declarations

Ethical approval Not required.

Competing interests The authors declare no competing interests.

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